IRON WITHIN RESEARCH

PT-141 — Certificates of Analysis

Every batch we have tested for this product, newest first.
Match the strength on your vial to the rows below and turn to that page.

PAGE	STRENGTH	BATCH / LOT
2–3	10mg, 36mg	10/9/2026 · Lot IWR-1010-PT141
4–7	10mg, 13mg, 85mg	8/2/2026 · Lot IWR-2607-PT14110L

Page numbers include this index page. Testing is per production batch; one batch can cover several vial sizes.

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0304

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
PT-141 — 10 mg

PASS

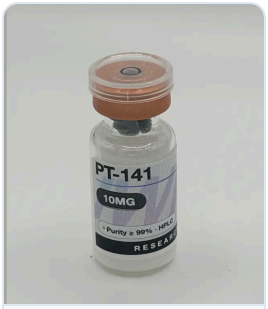
BATCH #	PURITY	MASS	IDENTITY
IWR-1010-PT141	99.34%	10.36 mg	Confirmed

COA #:	SBA-26-0304	Method:	Full QC Panel
Lot Number:	IWR-1010-PT141	Analysis Date:	2026-10-09
Labeled Content:	10 mg	Date Received:	2026-10-06
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	PT-141	Confirmed
Purity (HPLC)	>= 98.0%	99.34%
Content (per vial)	Report only	10.36 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.04 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits



SAMPLE AS RECEIVED

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

Kasey Malganius

Kasey Malganius
Lab Director
2026-10-09

COA #:	**SBA-26-0304**
Access Code:	**JF4LBMX4**
Analyst:	**A. Joseph**
Verify:	summitbioanalytics.com/verify
Issued:	2026-10-09

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

Horizon Analytical

Certificate of Analysis

Lot Number: IWR-6170177-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 04/21/2026
Analysis Conducted: 04/16/2026
Searchable via: horizonanalytical.com

Compound:	PT-141
Lot:	IWR-6170177-P
Appearance:	White Lyophilized Powder

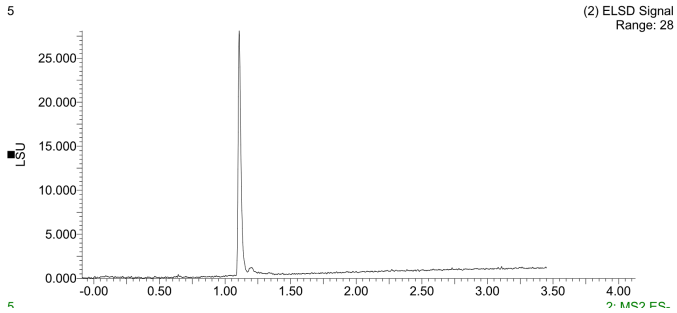
CAS:	189691-06-3
Formula:	$C_{50}H_{68}N_{14}O_{10}$
Mol Weight:	~1025.2 g/mol

Pubchem CID: 9941379

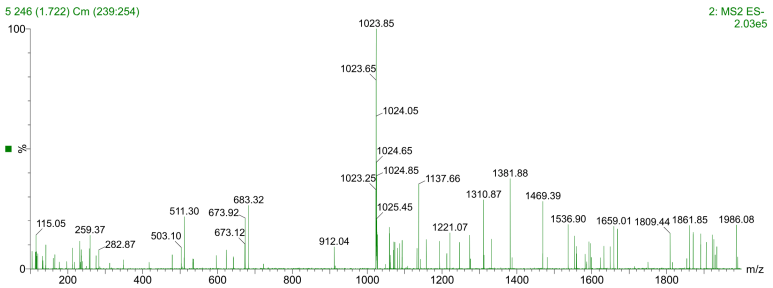
Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	PT-141	PT-141	
Quantity:	10mg	9.85mg	
Purity:	>98%	99.35%	

UPLC



MS



Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

Horizon Analytical

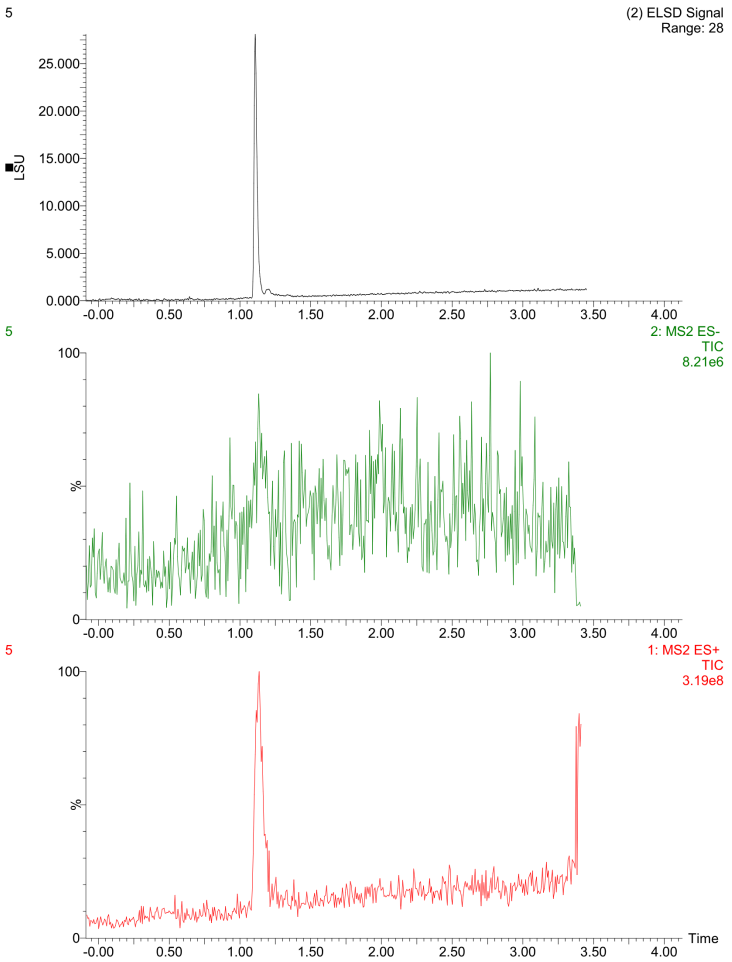
Certificate of Analysis

Lot Number: IWR-6170177-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

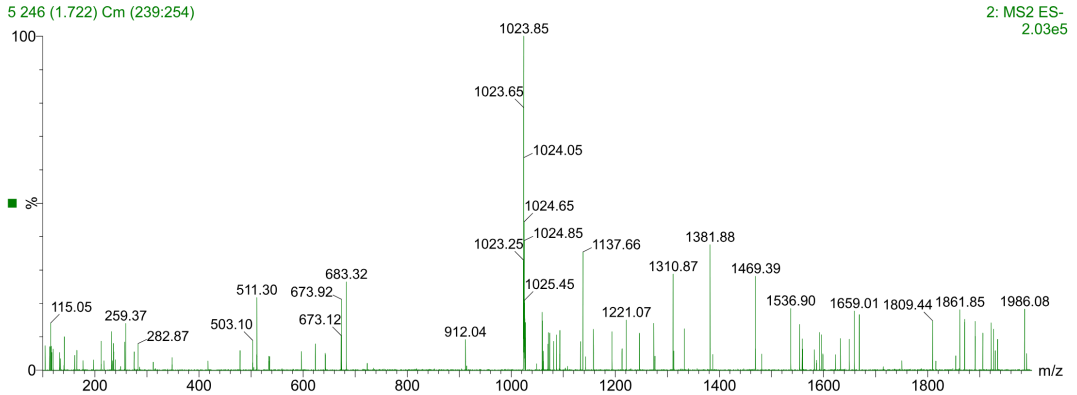
Received Date: 04/21/2026
Analysis Conducted: 04/16/2026
Searchable via: horizonanalytical.com

PT-141 (10mg) • Pubchem CID: 9941379

Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0023

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
PT-141 — 10 mg

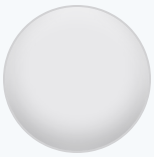
PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-2607-PT14110	99.74%	12.13 mg	Confirmed

COA #:	SBA-26-0023	Method:	Full QC Panel
Lot Number:	IWR-2607-PT14110	Analysis Date:	2026-08-02
Labeled Content:	10 mg	Date Received:	2026-07-29
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- Identity
- Purity
- Endotoxins
- Fentanyl Free



VIAL CAP
White
#E8E8EA

CLOSURE AS RECEIVED

Test	Specification	Result
Identity	PT-141	Confirmed
Purity (HPLC)	>= 98.0%	99.74%
Content (gross)	Report only	12.13 mg
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.07 EU/mL

Test	Specification	Result
Sterility Screen (rapid PCR)	No growth	**No growth**
Heavy Metals (ICP-MS)	ICH Q3D limits	**All analytes within limits**

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.



Kasey Malganius
Lab Director
2026-08-02

COA #:	**SBA-26-0023**
Access Code:	**JD7E6XED**
Analyst:	**R. Whitfield**
Verify:	summitbioanalytics.com/verify
Issued:	2026-08-02

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.